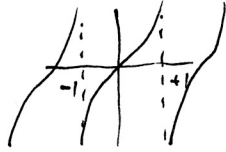


Problem 2. Find the Fourier Series (FS) of the periodic function $x(t)$.

PJ-1

2.1) $x(t) = \frac{\pi}{2} t^3$; $-1 < t < 1$



$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{j k \omega_0 t} ; \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j k \omega_0 t} dt$$

$$a_k = \frac{1}{2} \int_{-1}^{+1} \frac{\pi}{2} t^3 e^{-j k \omega_0 t} dt$$

$$= \frac{\pi}{4} \int_{-1}^{+1} t^3 e^{-j k \omega_0 t} dt$$

$e^{j\theta} = \cos\theta + j\sin\theta$; Euler Formulas, $e^{-j\theta} = \cos\theta - j\sin\theta$

$$= \frac{\pi}{4} \int_{-1}^{+1} t^3 (\cos(k\omega_0 t) - j\sin(k\omega_0 t)) dt$$

$$= \frac{\pi}{4} \left[\int_{-1}^{+1} t^3 \cos(k\omega_0 t) dt - j \int_{-1}^{+1} t^3 \sin(k\omega_0 t) dt \right]$$

Integration By parts: $\int u dv = uv - \int v du$

$$\int_{-1}^{+1} t^3 \cos(k\omega_0 t) dt$$

$$u = t^3, \quad dv = \cos(k\omega_0 t) dt$$

$$du = 3t^2 dt, \quad v = \int \cos(k\omega_0 t) dt = \frac{1}{k\omega_0} \sin(k\omega_0 t)$$

$$\int_{-1}^{+1} t^3 \cos(k\omega_0 t) dt = \left. t^3 \frac{\sin(k\omega_0 t)}{k\omega_0} \right|_{-1}^{+1} - \int_{-1}^{+1} \frac{\sin(k\omega_0 t)}{k\omega_0} \cdot 3t^2 dt$$

$$\left[\int t^n \cos(k\omega_0 t) dt = t^n \frac{\sin(k\omega_0 t)}{k\omega_0} - \int \frac{\sin(k\omega_0 t)}{k\omega_0} n t^{n-1} dt \right]$$

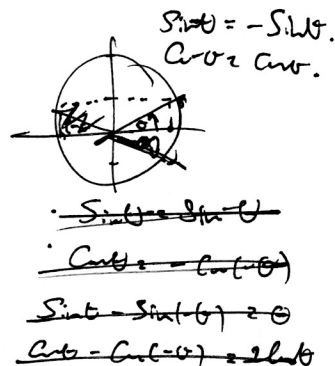
$$\left[\int t^n \sin(k\omega_0 t) dt = t^n \frac{(-\cos(k\omega_0 t))}{k\omega_0} + \int \frac{\cos(k\omega_0 t)}{k\omega_0} n t^{n-1} dt \right]$$

$$\int \frac{\sin(k\omega_0 t)}{k\omega_0} \cdot 3t^2 dt = \frac{3}{k\omega_0} \left(\frac{-t^2 \cos(k\omega_0 t)}{k\omega_0} + \frac{2}{k\omega_0} \int \cos(k\omega_0 t) t dt \right)$$

$$\int \cos(k\omega_0 t) t dt = t \frac{\sin(k\omega_0 t)}{k\omega_0} - \frac{1}{k\omega_0} \int \sin(k\omega_0 t) dt \cdot \frac{1}{k\omega_0}$$

$$= t \frac{\sin(k\omega_0 t)}{k\omega_0} + \frac{1}{(k\omega_0)^2} \cos(k\omega_0 t)$$

$$\int \frac{\sin(k\omega_0 t)}{k\omega_0} 3t^2 dt = \frac{-3t^2 \cos(k\omega_0 t)}{(k\omega_0)^2} + \frac{6}{(k\omega_0)^2} \left(t \frac{\sin(k\omega_0 t)}{k\omega_0} + \frac{\cos(k\omega_0 t)}{(k\omega_0)^2} \right)$$



$$\int \frac{\sin(h\omega t)}{h\omega} 3t^2 dt = -\frac{3t^2}{(h\omega)^2} \cos(h\omega t) + \frac{6t \sin(h\omega t)}{(h\omega)^3} + \frac{6 \cos(h\omega t)}{(h\omega)^4}$$

$$\therefore \int t^3 \cos(h\omega t) dt = \frac{t^3}{h\omega} \sin(h\omega t) + \frac{3t^2}{(h\omega)^2} \cos(h\omega t) - \frac{6t}{(h\omega)^3} \sin(h\omega t) - \frac{6}{(h\omega)^4} \cos(h\omega t)$$

$$\cos(\theta + \frac{\pi}{2}) = -\sin \theta$$

$$\int \cos \theta = \sin(\theta + \frac{\pi}{2}) \quad \text{integration by parts} \quad \frac{d}{dt} \cos(h\omega t) = -h\omega \sin(h\omega t) \quad dt = d(t + \text{const.}) \quad \frac{d}{dt} \sin(h\omega t) = h\omega \cos(h\omega t)$$

$$\therefore \int t^3 \sin(h\omega t) dt = -\frac{t^3}{h\omega} \cos(h\omega t) + \frac{3t^2}{(h\omega)^2} \sin(h\omega t) + \frac{6t}{(h\omega)^3} \cos(h\omega t) - \frac{6}{(h\omega)^4} \sin(h\omega t)$$

$$\therefore \int_{t=-1}^{t=1} t^3 \cos(h\omega t) dt = \frac{2}{h\omega} \sin(h\omega) + \frac{6}{(h\omega)^2} \cos(h\omega) - \frac{6 \cos(h\omega)}{(h\omega)^4} = \frac{28 \sin(h\omega)}{h\omega} - \frac{12 \sin(h\omega)}{(h\omega)^3} = 0$$

$$\int_{t=-1}^{t=1} t^3 \sin(h\omega t) dt = -\frac{2}{h\omega} \cos(h\omega) + \frac{12 \cos(h\omega)}{(h\omega)^3} - \frac{6 \sin(h\omega)}{(h\omega)^2} - \frac{12 \sin(h\omega)}{(h\omega)^4}$$

$$\therefore a_h = \frac{1}{4} \left\{ \int_{t=-1}^{t=1} t^3 \cos(h\omega t) dt - j \int_{t=-1}^{t=1} t^3 \sin(h\omega t) dt \right\}$$

$$= \frac{1}{4} \left\{ \frac{6 \cos(h\omega)}{(h\omega)^2} - \frac{6 \cos(h\omega)}{(h\omega)^4} + j \frac{2 \cos(h\omega)}{h\omega} - j \frac{12 \cos(h\omega)}{(h\omega)^3} \right\}$$

$$\therefore a_h = \frac{3\pi \cos(h\omega)}{2} \left(1 - \frac{1}{(h\omega)^2} \right) + j \frac{\pi \cos(h\omega)}{2h\omega} \left(1 - \frac{6}{(h\omega)^2} \right)$$

$$= \frac{\pi}{4} \left\{ (\cos(h\omega) (h\omega)^2 - 6 \cos(h\omega)) + j 2 \cos(h\omega) (h\omega)^3 - j 12 \cos(h\omega) (h\omega) / (h\omega)^4 \right\}$$

$$= \frac{3 \cos(h\omega) \pi}{2} - j \frac{4 \cos(h\omega) \pi}{2} + \frac{3 \pi (h\omega)^2 \cos(h\omega)}{2} + \frac{j \pi (h\omega)^3 \cos(h\omega)}{2 (h\omega)^4}$$

$$\therefore \int_{t=-1}^{t=1} t^3 \sin(h\omega t) dt = -\frac{2}{h\omega} \cos(h\omega) + \frac{6}{(h\omega)^2} \sin(h\omega) + \frac{12 \cos(h\omega)}{(h\omega)^3} - \frac{12 \sin(h\omega)}{(h\omega)^4}$$

$$\therefore a_h = \frac{\pi}{4} \left\{ 0 - j \left(\dots \right) \right\} = j \left\{ \frac{\pi \cos(h\omega)}{2 h\omega} - \frac{3\pi \sin(h\omega)}{2 (h\omega)^2} - \frac{3\pi \cos(h\omega)}{(h\omega)^3} + \frac{3\pi \sin(h\omega)}{(h\omega)^4} \right\}$$

$$= j \frac{\pi}{2 (h\omega)^4} \left\{ \pi \cos(h\omega) h^2 \omega^3 - 3\pi \sin(h\omega) h^2 \omega^2 - 6\pi \cos(h\omega) h \omega + 6\pi \sin(h\omega) \right\}$$

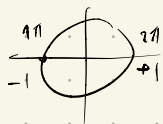
$$\therefore a_h = \frac{j}{2\pi^4 h^4} \left\{ \pi^4 h^3 \cos(h\omega) - 3\pi^3 h^2 \sin(h\omega) - 6\pi^2 h \cos(h\omega) + 6\pi \sin(h\omega) \right\}$$

$$\text{Case: } h = 0 : a_{h=0} = \frac{\pi}{4} \int_{-1}^1 t^3 e^{j0} dt = \frac{\pi}{4} \left. \frac{t^4}{4} \right|_{-1}^1 = 0$$

$$\therefore x_A = \sum_{h=-\infty}^{\infty} a_h e^{j h \omega t} ; a_h = \begin{cases} 0 & ; h = 0 \\ j \left(\frac{\pi^4 h^3 \cos(h\omega) - 3\pi^3 h^2 \sin(h\omega) - 6\pi^2 h \cos(h\omega) + 6\pi \sin(h\omega)}{2\pi^4 h^4} \right) & ; h \neq 0 \end{cases}$$

$$; \omega = 2\pi ; h \neq 0$$

for h is integer: $a_h = \begin{cases} 0 & ; h = 0 \end{cases}$



$$\begin{cases} j \frac{1}{2h} \left(\frac{e}{\pi^{2h}} - 1 \right) & ; h \text{ is odd} \\ j \frac{1}{2h} \left(1 - \frac{e}{\pi^{2h}} \right) & ; h \text{ is even} \end{cases}$$

~~#~~

2.2)

$$x(t) = \pi - t; \quad -\pi \leq t \leq \pi$$

Fourier Series: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkh\omega t}$; $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{(2\pi)} = 1$, $T = 2\pi$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jkh\omega t} dt$$

$$a_k = \frac{1}{2\pi} \int_{t=-\pi}^{\pi} (\pi - t) e^{-jkh\omega t} dt$$

Case: $h=0$

$$a_{h=0} = \frac{1}{2\pi} \int_{t=-\pi}^{\pi} (\pi - t) dt = \frac{-1}{2\pi} \int_{t=-\pi}^{\pi} (\pi - t) d(\pi - t) = \frac{-1}{2\pi} \left. \frac{(\pi - t)^2}{2} \right|_{t=-\pi}^{+\pi}$$

$$= \frac{-1}{2\pi} \left(0 - \frac{4\pi^2}{2} \right) = \pi \quad \therefore a_0 = \pi$$

Case: $h \neq 0$

$$a_k = \frac{1}{2\pi} \left(\int_{t=-\pi}^{\pi} \pi e^{-jht} dt + \int_{t=-\pi}^{\pi} -t e^{-jht} dt \right) = \frac{1}{2\pi} \int_{t=-\pi}^{\pi} (\pi - t) (\cos(ht) - j \sin(ht)) dt$$

$$= \frac{1}{2\pi} \left\{ \int_{t=-\pi}^{\pi} \pi \cos(ht) dt - j \int_{t=-\pi}^{\pi} \pi \sin(ht) dt - \int_{t=-\pi}^{\pi} t \cos(ht) dt + j \int_{t=-\pi}^{\pi} t \sin(ht) dt \right\}$$

By parts

$$= \frac{1}{2\pi} \left\{ \frac{\pi \sin(ht)}{h} + j \frac{\pi \cos(ht)}{h} - t \frac{\sin(ht)}{h} + \frac{1}{h} \frac{(-\cos(ht))}{h} + j \left[t \frac{(-\cos(ht))}{h} - \frac{1}{h} \frac{\sin(ht)}{h} \right] \right\} \Big|_{t=-\pi}^{t=+\pi}$$

$$= \frac{1}{2\pi} \left\{ \frac{2\pi \sin(h\pi)}{h} - j \frac{2\pi \cos(h\pi)}{h} + j \frac{2 \sin(h\pi)}{h^2} \right\}$$

$$= \frac{\sin(h\pi)}{h} - j \frac{\cos(h\pi)}{h} + j \frac{\sin(h\pi)}{h^2 \pi}$$

$$\therefore \text{Fourier Series } x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkh\omega t} ; a_k = \begin{cases} \pi & ; h=0 \\ \frac{\sin(h\pi)}{h^2 \pi} + j \left(\frac{\sin(h\pi)}{h^2 \pi} - \frac{\cos(h\pi)}{h} \right) & ; h \neq 0 \end{cases}$$

For h is integer: $a_k = \begin{cases} 0 & ; h=0 \\ j/h & ; h \text{ is odd} \\ -j/h & ; h \text{ is even} \end{cases}$

Problem 2-3

$$x(t) = t^2 + \sin^3(\pi t) \quad ; \quad -1 \leq t \leq 1$$

Fourier Series: $x(t) = \sum_{h=-\infty}^{\infty} a_h e^{jh\omega t}$, $\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$

$$a_h = \frac{1}{T} \int_{(T)} x(t) e^{-jh\omega t} dt$$

$$a_h = \frac{1}{2} \int_{t=-1}^{+1} (t^2 + \sin^3(\pi t)) e^{-jh\omega t} dt$$

$$= \frac{1}{2} \int_{t=-1}^{+1} t^2 e^{-jh\omega t} dt + \frac{1}{2} \int_{t=-1}^{+1} \sin^3(\pi t) e^{-jh\omega t} dt \quad \left. \begin{array}{l} \sin 3\theta = 3\sin\theta - 4\sin^3\theta \\ \sin^3\theta = \frac{3\sin\theta}{4} - \frac{1\sin 3\theta}{4} \end{array} \right\}$$

$$= \frac{1}{2} \int t^2 e^{-jh\omega t} dt + \frac{1}{2} \int \left(\frac{3}{4} \sin(\pi t) - \frac{1}{4} \sin(3\pi t) \right) e^{-jh\omega t} dt \quad \left. \begin{array}{l} \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} \end{array} \right\}$$

$$= \frac{1}{2} \int t^2 e^{-jh\omega t} dt + \frac{1}{8} \int 3 \left(\frac{e^{j\pi t}}{2j} - \frac{e^{-j\pi t}}{2j} \right) e^{-jh\omega t} - \left(\frac{e^{j3\pi t}}{2j} - \frac{e^{-j3\pi t}}{2j} \right) e^{-jh\omega t} dt$$

$$= \frac{1}{2} \int t^2 e^{-jh\omega t} dt + \frac{1}{16j} \int \left(3 e^{j\pi t(1-h)} - 3 e^{j\pi t(1+h)} \right) - \left(e^{j\pi t(3-h)} - e^{j\pi t(3+h)} \right) dt$$

$$= \frac{1}{2} \int t^2 e^{-jh\omega t} dt + \frac{3}{16j} \int e^{j\pi(1-h)t} dt - \frac{3}{16j} \int e^{j\pi(1+h)t} dt - \frac{1}{16j} \int e^{j\pi(3-h)t} dt + \frac{1}{16j} \int e^{j\pi(3+h)t} dt$$

$\left. \begin{array}{l} \text{Integration By parts} \\ \int t^n e^{-jh\omega t} dt \end{array} \right\}$

$$u = t^n, \quad dv = e^{-jh\omega t} dt$$

$$du = n t^{n-1} dt$$

$$v = \int e^{-jh\omega t} dt \frac{(-jh\omega)}{(-jh\omega)} = \frac{e^{-jh\omega t}}{-jh\omega}$$

$$\therefore \int t^n e^{-jh\omega t} dt = t^n \frac{e^{-jh\omega t}}{-jh\omega} - \int \frac{e^{-jh\omega t}}{-jh\omega} \cdot n t^{n-1} dt$$

$$= \frac{1}{2} t^2 \frac{e^{-jh\omega t}}{-jh\omega} - \frac{1}{2} \int \frac{e^{-jh\omega t}}{-jh\omega} \cdot 2t dt + \dots$$

$$= \frac{t^2}{2} \frac{e^{-jh\omega t}}{-jh\omega} - \frac{1}{jh\omega} \int \underbrace{e^{-jh\omega t} \cdot t}_{\text{By parts}} dt + \dots$$

$$= \frac{t^2}{2} \frac{e^{-j\omega t}}{-j\omega\pi} - \frac{1}{-j\omega\pi} \left(\frac{t e^{-j\omega t}}{-j\omega\pi} - \frac{1}{-j\omega\pi} \int e^{-j\omega t} dt \right) + \dots$$

$$= \frac{t^2}{2} \frac{e^{-j\omega t}}{(-j\omega\pi)} - \frac{1}{(-j\omega\pi)^2} t e^{-j\omega t} + \frac{e^{-j\omega t}}{(-j\omega\pi)^3} + \dots$$

$$= \frac{t^2}{2} \frac{e^{-j\omega t}}{(-j\omega\pi)} - t \frac{e^{-j\omega t}}{(-j\omega\pi)^2} + \frac{e^{-j\omega t}}{(-j\omega\pi)^3} + \frac{3}{16j} \int e^{j\pi(1-h)t} dt - \frac{3}{16j} \int e^{-j\pi(1+h)t} dt - \frac{1}{16j} \int e^{j\pi(3-h)t} dt + \frac{1}{16j} \int e^{-j\pi(3+h)t} dt$$

$$= \frac{t^2}{2} \frac{e^{-j\omega t}}{(-j\omega\pi)} \Big|_{t=1}^{+1} - t \frac{e^{-j\omega t}}{(-j\omega\pi)^2} \Big|_{t=1}^{+1} + \frac{e^{-j\omega t}}{(-j\omega\pi)^3} \Big|_{t=1}^{+1} + \frac{3}{16j} \frac{e^{j\pi(1-h)t}}{j\pi(1-h)} \Big|_{t=1}^{+1} - \frac{3}{16j} \frac{e^{-j\pi(1+h)t}}{(-j\pi(1+h))} \Big|_{t=1}^{+1} - \frac{1}{16j} \frac{e^{j\pi(3-h)t}}{j\pi(3-h)} \Big|_{t=1}^{+1} + \frac{1}{16j} \frac{e^{-j\pi(3+h)t}}{(-j\pi(3+h))} \Big|_{t=1}^{+1}$$

$$= \frac{1}{h\pi} \left(\frac{e^{j\omega} - e^{-j\omega}}{2j} \right) + \frac{2}{h^2\pi^2} \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right) - \frac{2}{h^3\pi^3} \left(\frac{e^{j\omega} - e^{-j\omega}}{2j} \right) + \frac{1}{16j} \frac{e^{j\pi(3-h)t}}{(-j\pi(3+h))} \Big|_{t=1}^{+1}$$

$$- \frac{3j}{8\pi(1-h)} \left(\frac{e^{j\pi(1-h)} - e^{-j\pi(1-h)}}{2j} \right) + \frac{3j}{8\pi(1+h)} \left(\frac{e^{j\pi(1+h)} - e^{-j\pi(1+h)}}{2j} \right)$$

$$+ \frac{j}{8\pi(3-h)} \left(\frac{e^{j\pi(3-h)} - e^{-j\pi(3-h)}}{2j} \right) - \frac{j}{8\pi(3+h)} \left(\frac{e^{j\pi(3+h)} - e^{-j\pi(3+h)}}{2j} \right)$$

$$= \frac{\text{Sinc}(h\pi)}{h\pi} + \frac{2}{h^2\pi^2} \text{Cos}(h\pi) - \frac{2}{h^3\pi^3} \text{Sinc}(h\pi) - j \frac{3}{8\pi(1-h)} \text{Sinc}(\pi(1-h)) + j \frac{3}{8\pi(1+h)} \text{Sinc}(\pi(1+h))$$

$$+ j \frac{1}{8\pi(3-h)} \text{Sinc}(\pi(3-h)) - j \frac{1}{8\pi(3+h)} \text{Sinc}(\pi(3+h))$$

$$\therefore a_h = \text{Sinc}(h\pi) + \frac{2}{h^2\pi^2} (\text{Cos}(h\pi) - \text{Sinc}(h\pi)) - j \frac{3}{8} \text{Sinc}(\pi(1-h)) + j \frac{3}{8} \text{Sinc}(\pi(1+h))$$

$$+ j \frac{1}{8} \text{Sinc}(\pi(3-h)) - j \frac{1}{8} \text{Sinc}(\pi(3+h))$$

Case $h=0$;

$$\begin{aligned}
 a_0 &= \frac{1}{2} \int_{-1}^{+1} (t^2 + \sin^3(\pi t)) dt = \frac{1}{2} \left[\frac{t^3}{3} \right]_{-1}^{+1} + \frac{1}{2} \int_{-1}^{+1} \sin^3(\pi t) dt = \frac{1}{3} + \frac{1}{2\pi} \int_{-1}^{+1} \sin^3(\pi t) d(\pi t) \\
 &= \frac{1}{3} + \frac{1}{2\pi} \int_{-1}^{+1} \frac{3}{4} \sin(\pi t) - \frac{1}{4} \sin(3\pi t) dt \\
 &= \frac{1}{3} - \frac{1}{4\pi} \frac{3}{8} \cos(\pi t) \Big|_{-1}^{+1} + \frac{1}{24\pi} \cos(3\pi t) \Big|_{-1}^{+1}
 \end{aligned}$$

$$\therefore a_0 = \frac{1}{3}$$

$$\therefore \text{Fourier Series: } x(t) = \sum_{h=-\infty}^{+\infty} a_h e^{j\pi h t}$$

$$\begin{aligned}
 a_h &= \begin{cases} 1/3 & ; h=0 \\ \frac{\text{Sinc}(h\pi)}{h^2\pi^2} + \frac{2}{h^2\pi^2} (\cos(h\pi) - \text{Sinc}(h\pi)) & ; h \neq 0 \\ -j \frac{3}{8} \text{Sinc}(\pi(1-h)) + j \frac{3}{8} \text{Sinc}(\pi(1+h)) \\ + j \frac{1}{8} \text{Sinc}(\pi(3-h)) - j \frac{1}{8} \text{Sinc}(\pi(3+h)) \end{cases}
 \end{aligned}$$

for h is integer

$$\therefore a_h = \begin{cases} 0 & ; h=0 \\ \frac{-2}{h^2\pi^2} & ; h \text{ is odd} \\ \frac{+2}{h^2\pi^2} & ; h \text{ is even} \end{cases}$$

Problem 4

Let $\mathcal{F}\{x(t)\} = X(j\omega) = \text{rect}[(\omega-1)/2]$ Find Fourier transform of

1.) $x(-2t+4)$

Time Scaling: $\mathcal{F}\{x(\alpha t)\} = \frac{1}{|\alpha|} X\left(\frac{j\omega}{\alpha}\right)$

$$\mathcal{F}\{x(-2t)\} = \frac{1}{2} X\left(\frac{j\omega}{-2}\right) = \frac{1}{2} \text{rect}\left[\frac{-\frac{\omega}{2}-1}{2}\right] = \frac{1}{2} \text{rect}\left[-\frac{\omega+2}{4}\right]$$

$\{ \text{rect is even f} \} \quad \text{rect}(x) = \text{rect}(-x)$

$$\therefore \mathcal{F}\{x(-2t)\} = \frac{1}{2} \text{rect}\left[\frac{\omega+2}{4}\right]$$

Time Shifting: $\mathcal{F}\{x(t-t_0)\} = X(j\omega) e^{-j\omega t_0}$

$$\therefore \mathcal{F}\{x(-2t+4)\} = \frac{1}{2} \text{rect}\left[\frac{\omega+2}{4}\right] e^{-j\omega(2)} \quad *$$

2) $(t-1)x(t-1)$

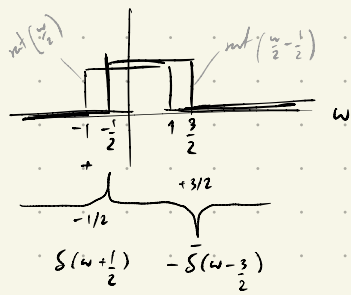
Time Shifting: $\mathcal{F}\{x(t-1)\} = X(j\omega) e^{-j\omega(1)}$
 $= \text{rect}\left[\frac{\omega-1}{2}\right] e^{-j\omega}$

O.A. in freq. $\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
 $= \int x(t) (-jt) e^{-j\omega t} dt$
 $= -j \int t x(t) e^{-j\omega t} dt$

$$\therefore \frac{d}{d\omega} X(j\omega) = -j \mathcal{F}\{t x(t)\}, \quad \mathcal{F}\{t x(t)\} = j \frac{d}{d\omega} X(j\omega) = j \frac{d}{d\omega} \mathcal{F}\{x(t)\}$$

$$\mathcal{F}\{t \cdot x(t)\} = j \frac{d}{d\omega} \mathcal{F}\{x(t)\}$$

$$= j \frac{d}{d\omega} \left(\text{rect}\left[\frac{\omega-1}{2}\right] e^{-j\omega} \right)$$



$$= j \left\{ \text{rect}\left[\frac{\omega-1}{2}\right] e^{-j\omega} (-j) + e^{-j\omega} \frac{d}{d\omega} \text{rect}\left[\frac{\omega-1}{2}\right] \right\}$$

$$= \text{rect}\left[\frac{\omega-1}{2}\right] e^{-j\omega} + j e^{-j\omega} \left(\delta\left(\omega+\frac{1}{2}\right) - \delta\left(\omega-\frac{3}{2}\right) \right) \quad *$$

$j \delta$ is Dirac delta function

$$3.) \mathcal{F}\left\{t \frac{d}{dt} x(t)\right\} = ?$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} (j\omega) d\omega$$

$$\underbrace{\frac{d}{dt} x(t)}_{f(t)} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \underbrace{j\omega X(j\omega)}_{f(\omega)} \underbrace{e^{j\omega t}}_{\text{kernel}} d\omega \quad \} \quad \mathcal{F}(f(t)) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} f(\omega) e^{j\omega t} d\omega$$

$$\therefore \mathcal{F}\left\{t \frac{d}{dt} x(t)\right\} = j\omega X(j\omega)$$

$$\mathcal{F}\left\{t \frac{d}{dt} x(t)\right\} = j \frac{d}{d\omega} \mathcal{F}\left\{\frac{d}{dt} x(t)\right\}$$

$$= j \frac{d}{d\omega} \left(j\omega X(j\omega) \right)$$

$$= j^2 \left(\omega \frac{d}{d\omega} X(j\omega) + X(j\omega) \right)$$

$$= -\omega \frac{d}{d\omega} \text{rect}\left[\frac{\omega-1}{2}\right] - \text{rect}\left[\frac{\omega-1}{2}\right]$$

$$= -\omega \left(\delta\left(\omega+\frac{1}{2}\right) - \delta\left(\omega-\frac{3}{2}\right) \right) - \text{rect}\left[\frac{\omega-1}{2}\right] \quad *$$

$j \delta$ is Dirac delta function

$$4.) \mathcal{F}\{x(2t-1)e^{-j2t}\} = ?$$

Time Scaling: $\mathcal{F}\{x(2t)\} = \frac{1}{2} X(j\frac{\omega}{2}) = \frac{1}{2} \text{rect}\left[\frac{\omega-2}{4}\right]$

Time Shifting: $\mathcal{F}\{x(2t-1)\} = \frac{1}{2} \text{rect}\left[\frac{\omega-2}{4}\right] e^{-j\omega(\frac{1}{2})}$

Frequency Shifting: $\mathcal{F}\{x(t)e^{j\omega_0 t}\} = X(j(\omega-\omega_0))$

$$\begin{aligned} \mathcal{F}\{x(2t-1)e^{j(-2)t}\} &= X(j(\omega-(-2))) \\ &= \frac{1}{2} \text{rect}\left[\frac{\omega+2-2}{4}\right] e^{-j(\omega+2)\frac{1}{2}} \\ &= \frac{1}{2} \text{rect}\left[\frac{\omega}{4}\right] e^{-j(\frac{\omega+2}{2})} \end{aligned}$$

Sol-II

$$\begin{aligned} \mathcal{F}\{x(2t-1)e^{-j2t}\} &= \int_{t=-\infty}^{+\infty} x(2t-1)e^{-j2t} \cdot e^{-j\omega t} dt \\ &= \int_{t=-\infty}^{+\infty} x(2t-1)e^{-j(\omega+2)t} dt \\ &\quad \left\{ \begin{array}{l} t' = 2t-1, \quad t = \frac{t'+1}{2}, \quad dt = \frac{1}{2} dt' \end{array} \right. \\ &= \int_{t'=-\infty}^{+\infty} x(t')e^{-j(\omega+2)(\frac{t'+1}{2})} \frac{1}{2} dt' \\ &= \frac{1}{2} \int_{t'=-\infty}^{+\infty} x(t')e^{-j(\omega+2)\frac{t'}{2}} e^{-j(\omega+2)\frac{1}{2}} dt' \\ &= \frac{1}{2} e^{-j(\frac{\omega+2}{2})} \int_{t'=-\infty}^{+\infty} x(t')e^{-j(\frac{\omega+2}{2})t'} dt' \\ &= \frac{1}{2} e^{-j(\frac{\omega+2}{2})} X(j(\frac{\omega+2}{2})) \\ &= \frac{1}{2} e^{-j(\frac{\omega+2}{2})} \text{rect}\left[\frac{(\frac{\omega+2}{2})-1}{2}\right] \end{aligned}$$

$$\therefore \mathcal{F}\{x(2t-1)e^{-j2t}\} = \frac{1}{2} e^{-j(\frac{\omega+2}{2})} \text{rect}\left[\frac{\omega}{4}\right]$$

$$5.) \mathcal{F}\{x(t) * x(t-1)\} = ?$$

$$\mathcal{F}\{x(t)\} = X(j\omega)$$

$$\mathcal{F}\{x(t-1)\} = X(j\omega) e^{-j\omega(1)}$$

$$\begin{aligned} \text{Convolution: } \mathcal{F}\{x(t) * x(t-1)\} &= X(j\omega) \cdot X(j\omega) e^{-j\omega} \\ &= x(t) \left[\frac{\omega-1}{2} \right] e^{-j\omega} \quad \# \end{aligned}$$